MATH3881/5881: Statistical Machine Learning Theory

Week 1: Supplement: Geometry of the Maximum-Margin Classifier

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Optional reading. This supplement is optional material for students who want a deeper geometric understanding of support vector machines. It does not introduce any new examinable content beyond Section 1.7 of the Week 1 lecture notes; students may safely skip it.

This short supplement extends Section 1.7 (Support Vector Machines) of the Week 1 lecture notes. It explains the choice of ± 1 labels, states the SVM's goal in plain words, derives the margin formula $2/\|w\|$, shows how the soft-margin SVM collapses to ERM with hinge loss, and closes with a comparison to logistic regression.

1. Why labels $y \in \{-1, +1\}$?

Throughout Week 1 we used $y \in \{0, 1\}$ for binary classification: it matches the Bernoulli probability model behind binary cross-entropy. For SVMs we switch convention to $y \in \{-1, +1\}$. The reason is purely notational: with this encoding, the product

$$y^{(i)} \cdot \left(w^\top x^{(i)} + b\right)$$

is *positive* exactly when the prediction is correct, and its magnitude measures how far the point sits on the correct side of the hyperplane $\{x : w^\top x + b = 0\}$. The single inequality $y^{(i)}(w^\top x^{(i)} + b) \geq 1$ then cleanly says "the point is correctly classified with at least a unit margin", whether $y^{(i)}$ is $+1$ or -1. With $y \in \{0, 1\}$ one would need to write two cases, one for each label. The choice is cosmetic: it is the same model on the same data, just with the relabelling $\tilde{y} = 2y - 1$.

2. What does SVM do?

For a linearly separable binary classification problem, *many* hyperplanes will correctly separate the two classes; any of them gives zero training error. The SVM picks a specific one, based on a clean geometric criterion: among all linear classifiers that separate the data, choose the one whose decision boundary lies as far as possible from the nearest training points.

The perpendicular distance from the decision boundary to the closest training point (on either side) is called the **margin**. A boundary with a larger margin is more robust: small perturbations to the inputs are less likely to push a point to the wrong side. The SVM's goal can therefore be stated in one line:

Among all hyperplanes that separate the data, find the one with the largest margin.

The training points that sit closest to the boundary, that is, *on* the two parallel margin boundaries shown in Figure 1.1 below, have a special name: they are called the **support vectors**. A key fact about the SVM solution is that it depends *only* on these support vectors: every other training point can be moved or removed without changing the solution. The name "support vector machine" reflects this picture: a handful of points "support" the position of the decision boundary, like pillars supporting a beam.

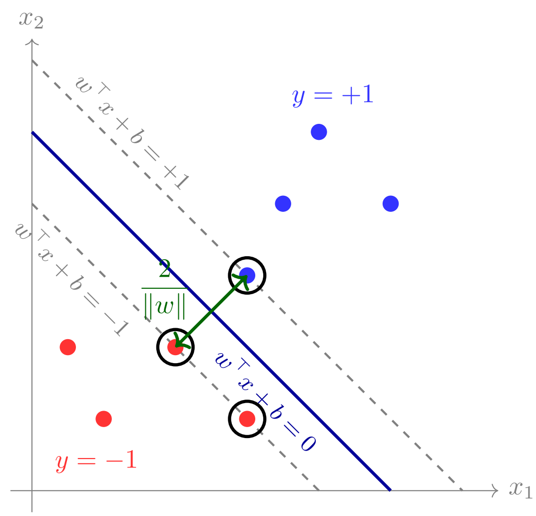

Figure 1.1: The maximum-margin classifier in $\mathbb{R}^2$. Solid blue: the decision boundary $w^\top x + b = 0$. Dashed grey: the parallel margin boundaries at $w^\top x + b = \pm 1$. Circled points are the *support vectors*: the only points whose removal would change the optimal hyperplane. All other points lie strictly inside their correct half-space and have no influence on the solution.

3. The geometric margin is $2/\|w\|$

The decision boundary $\{x : w^\top x + b = 0\}$ is a hyperplane in $\mathbb{R}^p$ with unit normal $w/\|w\|$. Let $x_\star$ be any point on the hyperplane, so that $w^\top x_\star + b = 0$. The (signed) perpendicular distance from a point $x_0 \in \mathbb{R}^p$ to the hyperplane is the projection of $x_0 - x_\star$ onto the unit normal:

$$d(x_0) \;=\; \frac{w^\top (x_0 - x_\star)}{\|w\|} \;=\; \frac{w^\top x_0 - w^\top x_\star}{\|w\|} \;=\; \frac{w^\top x_0 + b}{\|w\|}. \tag{1.1}$$

Now rescale (w, b) so that the points closest to the boundary on each side satisfy $|w^\top x^{(i)} + b| = 1$. (This is always possible: we can multiply both w and b by any positive scalar without changing the hyperplane.) Substituting into (1.1), the closest points lie at distance $1/\|w\|$ on each side, and the two parallel *margin boundaries* $w^\top x + b = +1$ and $w^\top x + b = -1$ are separated by

$$\boxed{\text{margin} \;=\; \frac{2}{\|w\|}.} \tag{1.2}$$

Maximising the margin is therefore equivalent to *minimising* $\|w\|$, or, more conveniently, $\frac{1}{2}\|w\|^2$ (a convex quadratic), subject to the constraints $y^{(i)}(w^\top x^{(i)} + b) \geq 1$. This is exactly the hard-margin SVM problem from §1.7 of the Week 1 notes.

4. From slack variables to hinge loss

The soft-margin SVM minimises

$$\tfrac{1}{2}\|w\|^2 + c\sum_{i=1}^{n}\xi_i \quad \text{subject to} \quad y^{(i)}(w^\top x^{(i)} + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \tag{1.3}$$

for some $c > 0$. The slack variables ξ_i are auxiliary: the objective is monotonically increasing in each ξ_i, so at the optimum each ξ_i takes its *smallest* permissible value. Combining the two constraints,

$$\xi_i \;\geq\; \max\bigl(0,\; 1 - y^{(i)}(w^\top x^{(i)} + b)\bigr),$$

so the minimum permissible slack is

$$\xi_i^\star \;=\; \max\bigl(0,\; 1 - y^{(i)}(w^\top x^{(i)} + b)\bigr). \tag{1.4}$$

This is exactly the *hinge loss* $\ell\bigl(y^{(i)}, \hat{y}^{(i)}\bigr) = \max\bigl(0, 1 - y^{(i)}\hat{y}^{(i)}\bigr)$ evaluated at the raw score $\hat{y}^{(i)} = w^\top x^{(i)} + b$. Substituting back into (1.3):

$$\tfrac{1}{2}\|w\|^2 + c\sum_{i=1}^{n}\max\bigl(0,\; 1 - y^{(i)}(w^\top x^{(i)} + b)\bigr).$$

Dividing through by nc and writing $\lambda = 1/(2nc)$ recovers the ERM form

$$\min_{w,b}\; \frac{1}{n}\sum_{i=1}^{n}\max\bigl(0,\; 1 - y^{(i)}(w^\top x^{(i)} + b)\bigr) \;+\; \lambda\,\|w\|^2.$$

So **the soft-margin SVM is exactly ERM with hinge loss and $\mathbf{L_2}$ regularisation on $\boldsymbol{w}$**. The slack-and-constraints story and the ERM-with-hinge-loss story describe the same optimisation problem.

Remark 1.0.1

A subtle but important point: the SVM uses the raw score $\hat{y}^{(i)} = w^\top x^{(i)} + b \in \mathbb{R}$, not a transformed probability. This is a different use of the symbol $\hat{y}$ than in the logistic-regression sections of Week 1, where $\hat{y} = \sigma(\theta^\top \tilde{x}) \in (0, 1)$ is a probability. The two are related: applying the sigmoid to the raw score gives a (heuristic) probability, but SVMs do not produce calibrated probabilities directly.

5. SVM vs. logistic regression: same family, different loss

Recall from Week 1 that logistic regression also produces a linear decision boundary: at the default threshold $\tau = 0.5$, the boundary is $\{x : w^\top x + b = 0\}$, the exact same form as the SVM boundary. So the SVM and logistic regression both live in the family of linear classifiers; they differ only in *which* hyperplane in that family they pick. The choice is dictated by the loss function used to fit (w, b).

Writing $z = y(w^\top x + b)$ for the *signed margin* of an example (positive when correct, negative when wrong), the two methods minimise different convex surrogates of the 0/1 loss:

$$\ell_{\text{hinge}}(z) = \max(0,\, 1 - z), \qquad \ell_{\text{logistic}}(z) = \log\bigl(1 + e^{-z}\bigr). \tag{1.5}$$

Both functions are convex, both upper-bound the 0/1 loss, and both push correctly-classified points (large z) toward zero loss while heavily penalising the wrong side. The key difference is the shape:

- **Hinge** has a kink at $z = 1$ and is *exactly zero* for $z \geq 1$. Points safely beyond the margin contribute nothing.
- **Logistic** is smooth and *strictly positive* everywhere. Every point contributes something, but exponentially less as z grows.

Two consequences follow directly:

- *Sparsity.* The flat region of the hinge loss is what makes SVM's solution depend only on the support vectors. Logistic regression, with its strictly positive loss, uses every training point to determine $\hat{w}$.
- *Confidence.* Composing the logistic-regression score $w^\top x + b$ with the sigmoid yields a calibrated probability $\hat{y} \in (0, 1)$. The SVM score is unbounded and is not, on its own, a probability.

A striking deep connection: on linearly separable data, L$_{\mathbf{2}}$-regularised logistic regression with $\boldsymbol{\lambda \to 0}$ converges, in the direction $\hat{\boldsymbol{w}}/\|\hat{\boldsymbol{w}}\|$, to the maximum-margin SVM solution. So in a precise limit, logistic regression *becomes* the SVM. SVM and logistic regression are not just analogous; they are two points on the same family of regularised ERM problems, distinguished only by the convex surrogate in (1.5).